

Joint Time-Frequency Offset Detections Using the Linear Canonical Transform

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Abstract—In this paper, a method to detect joint time-frequency offset is proposed based on the linear canonical transform. According to the fundamental properties of this transform, a parametric correlation is introduced, which can be regarded as the generalization of methods based on correlation and other transforms. Besides, its maximum can be identified with a line structure in the time-frequency plane. With the advantage of parameters' freedom in the linear canonical transform, some lines containing the time delay and frequency offset in the measured signals can be intersected in the time-frequency plane. And the intersected point is exactly joint time-frequency offset needed to detect out. The theoretical and practical aspects of this detection method are discussed in the paper.

Index Terms—Parametric correlation, time delay, frequency offset, linear canonical transform

I. INTRODUCTION

Time delay and frequency offset are both important and fundamental issues in the signal processing. It is applied in many areas such as radar, sonar, geophysics, biomedicine and ultrasonic imaging [1]. For example, in the radar detection, the echo signal is measured with a time delay compared with the original one. Once the time delay is detected, the distance between the receiver and object is easily obtained. Frequency offset is mainly influenced by Doppler effect, existing in many situations as well. Besides the single time delay or frequency shift, joint time delay and frequency shift (here we denote it as joint time-frequency offset) also influence how to decode the message properly in some applications. Thus, the detection of both time delay and frequency offset plays an important role in practical applications.

For single time delay detection, correlation operator is a conventional method, which is a classic way to measure the level of similarity for two signals [2]. Since auto-correlation function of a signal is even and achieves its maximum at origin point, time delay is easily detected by finding the maximum values of correlation for the model signal and its delayed version. For single frequency offset detection, it can be realized in a similar way by using correlation method based on Fourier transform (FT) [3]. Based on the property of frequency shift of FT, the frequency offset can be detected by searching the maximum of correlation between the FT of model signal and its frequency offset version. Although joint time-frequency

offset can be detected using the above two steps, the error will be large based on the existence of both time delay and frequency offset. The performance of time/frequency offset detection will descend because of the effect of frequency/time offset. To solve this problem, we propose a correlation method based on the linear canonical transform (LCT) to detect the joint time-frequency offset. Moreover, this technique is a generalization for the frame of correlation based on some transforms such as FT (called classical correlation) and the fractional Fourier transform (FRFT) (called α modulus correlation) [3].

The paper is mainly organized as follows. In Section 2, some basic knowledge about the LCT and correlation function is given briefly. We recall some basic properties of LCT used in this paper as well. In Section 3, the idea of parametric correlation are discussed based on the LCT. Some simulations are stated in the Section 4 to verify this approach. And the conclusion is in the Section 5.

II. PRELIMINARIES

In this Section, we mainly talk about LCT and conventional correlation function used in the sequel of this paper. Some fundamental properties are discussed as well.

A. Linear canonical transform

LCT was introduced in the 1970s, which is an important integral transform of four parameters with a determinant constraint [4]. It encompasses a number of transforms important in DSP and optical system modeling. The well-known signal processing operations such as FT, FRFT, the Fresnel transform (FST) and scaling operations are its special cases [5]. The LCT of a signal $x(t)$ with parameter matrix A is defined as

$$X^A(u) = LCT_x^A(u) = \int_{-\infty}^{+\infty} x(t)K_A(t, u)dt, \quad (1)$$

where $K_A(t, u)$ is kernel function. It is

$$K_A(t, u) = \begin{cases} \frac{1}{\sqrt{j2\pi b}} e^{j(\frac{du^2}{2b} - \frac{ut}{b} + \frac{at^2}{2b})}, & b \neq 0 \\ \sqrt{d} e^{j\frac{cd}{2}u^2} x(du), & b = 0 \end{cases} \quad (2)$$

with parameter matrix $A = (a, b; c, d)$ and the parameters a, b, c, d are real numbers satisfying $ad - bc = 1$, i.e., $\det(A) = 1$. When $A = (0, 1; -1, 0)$ and $A =$

$(\cos \alpha, \sin \alpha; -\sin \alpha, \cos \alpha)$, respectively, LCT reduces to the FT and FRFT, respectively. Obviously, the LCT of a signal is essentially a chirp signal multiplication when $b = 0$, and it is no interest to our research. Therefore, only $b \neq 0$ are chosen in this paper. In addition, here and in what follows the integral are taken over $(-\infty, \infty)$ if unspecified.

For analysing signals in joint time-frequency offset, it is necessary to consider two essential properties of LCT. Let $x_{t_d}(t) = x(t - t_d)$ denote time delay, $x_{f_d}(t) = x(t)e^{j2\pi f_d t}$ mean the frequency offset, and $x_{t_d f_d}(t) = x(t - t_d)e^{j2\pi f_d t}$ be joint time-frequency offset of $x(t)$, respectively. The two important properties of LCT are listed as follows [6]:

$$LCT_{x_{t_d}}^A(u) = e^{j(cut_d - \frac{1}{2}act_d^2)} L_x^A(u - at_d), \quad (3)$$

$$LCT_{x_{f_d}}^A(u) = e^{j(df_d u - \frac{1}{2}bdf_d^2)} L_x^A(u - bf_d). \quad (4)$$

Thus, for $x_{t_d f_d}(t)$, we have

$$|LCT_{x_{t_d f_d}}^A(u)| = |LCT_x^A(u - at_d - bf_d)|. \quad (5)$$

From Eq.5, the LCT spectrums of $x_{t_d f_d}(t)$ and $x(t)$ are almost the same except a shift in LCT domain. The shift contains the time delay t_d and frequency offset f_d .

B. Correlation function

For two finite energy signals $x(t)$ and $y(t)$, the correlation function is defined as [2]

$$C_{x,y}(t) = \int x(\tau)y(\tau - t)d\tau. \quad (6)$$

When $y(t) = x(t)$, the correlation is called auto-correlation of signal $x(t)$. It describes the similarity between $x(\tau)$ and $x(\tau - t)$. Since the correlation $C_{x,y}(t) = C_{y,x}(-t)$, the autocorrelation $C_{x,x}(t)$ is an even function and its maximum equals to the energy of $x(t)$ achieving at $t = 0$.

Based on the concept of correlation and its properties, time delay can be detected by correlating the model signal $x(t)$ and the delay signal $x_{t_d}(t)$, which is given by

$$C_{x,x_{t_d}}(t) = \int x(\tau)x_{t_d}(\tau - t)d\tau \quad (7)$$

and the time delay is

$$t_d = \max C_{x,x_{t_d}}. \quad (8)$$

For the frequency offset detection, similarly, it can be detected by

$$f_d = \max C_{X,X_{f_d}} \quad (9)$$

where X, X_{f_d} are the FT of signal $x(t), x_{f_d}(t)$, respectively.

III. PARAMETRIC CORRELATOR

For the signal with joint time-frequency offset, the detection is both for time delay and frequency offset. If a transform domain is not only about time, but also about frequency, it will contribute to the detection for joint time-frequency offset. Some transform domains such as FRFT domain and LCT domain are indeed between time and frequency, and they have free parameters compared with FT. Techniques based on FRFT are used in [3] [7]. However, LCT is a generalization of FRFT, and the freedom of parameters is much larger than FRFT. LCT not only contains on rotation of signal, but also about changing of scale in time-frequency analysis. Based on these advantages of LCT, a new parametric correlation method is proposed in this paper.

A. Parametric correlator

Definition 1: A parametric correlation function using the LCT for the given signal $x(t)$ and $y(t)$, is defined as

$$C_{x,y}^A(t) = C_{|LCT_x^A|, |LCT_y^A|}(t). \quad (10)$$

Compared with the conventional correlation, Eq. 10 uses the LCT spectrums of $x(t)$ and $y(t)$. Because of Eq. 3-5 and the properties of common correlation having, a parameter u^A including joint time-frequency offset can be detected, which is

$$u^A = at_d + bf_d = \max C_{|LCT_x^A|, |LCT_{x_{t_d f_d}}^A|}. \quad (11)$$

Especially, when $A = (0, 1; -1, 0)$ this parametric correlation reduces to the classical correlation; when $A = (\cos \alpha, \sin \alpha; -\sin \alpha, \cos \alpha)$, it turns to the α -modulus correlation [3].

If parameters t_d, f_d in Eq. 11 want to be solved, two equations are needed at least. Choosing two pairs of proper parameters (a, b) , two linearly independent equations are obtained, from which the joint time-frequency offset can be obtained directly by solving a equation set. For α -modulated correlator, there is only a parameter α , but this method contains two parameters (a, b) , which is a generalization of the former.

According to the equation derived in Eq. 11, n pairs of different parameters are given. The equations are listed as follows:

$$u^{A_i} = a_i t_d + b_i f_d = \max C_{|LCT_x^{A_i}|, |LCT_{x_{t_d f_d}}^{A_i}|} \quad (12)$$

where $i = 1, 2, \dots, n$. In this way, C_n^2 possible values (t_d, f_d) may be detected. The joint time-frequency offset can be obtained using some statistic techniques.

B. The choice of parameters

Before using this method to detect the time and frequency offset, some aspects, especially about the parameters of LCT should be considered firstly. Since the kernel function is a chirp signal, LCT is benefit for linear frequency modulated (LFM) signal processing. A LFM signal $s(t)$ is

$$s(t) = A_0 e^{j(\pi k t^2 + 2\pi f_0 t)} \quad (13)$$

where A_0, k, f_0 denote amplitude, modulated frequency and center frequency, respectively. Then the LCT of $s(t)$ is

$$\begin{aligned} S^A(u) &= \int s(t) K_A(t, u) dt \\ &= \frac{A_0}{\sqrt{j2\pi b}} \int e^{j(\pi k t^2 + 2\pi f_0 t)} e^{j(\frac{du^2}{2b} - \frac{ut}{b} + \frac{at^2}{2b})} dt \\ &= \frac{A_0}{\sqrt{j2\pi b}} e^{j\frac{d}{2b}u^2} \int e^{j(\pi k + \frac{a}{2b})t^2} e^{-j(\frac{u}{b} - 2\pi f_0)t} dt \end{aligned} \quad (14)$$

If $\pi k + a/2b = 0$, then Eq. 14 can be simplified to

$$\begin{aligned} S^A(u) &= \frac{A_0}{\sqrt{j2\pi b}} e^{j\frac{d}{2b}u^2} \int e^{-j(\frac{u}{b} - 2\pi f_0)t} dt \\ &= \frac{A_0}{\sqrt{j2\pi b}} e^{j\frac{d}{2b}u^2} \delta\left(\frac{u}{b} - 2\pi f_0\right) \end{aligned} \quad (15)$$

In the above analysis, the amplitude spectrum of $S^A(u)$ is a impulse in the LCT domain on the condition that $a/b = -2\pi k$. For FRFT, LFM signal is concentrated when $\alpha = -\text{arccot}(2\pi k)$. Such LCT domains with the same value a/b as well as their associated FRFT domain are referred to essentially equivalent domains [8]. And if a signal has a compact support in a certain LCT domain, then the signal will also have compact support in all essentially equivalent domains.

For digital processing of LCT and FRFT, there are a lot of algorithm existed [9]–[14]. In this paper, we mainly refer to these two algorithms [11], [12].

IV. SIMULATIONS

In this Section, we mainly use some simulations to discuss the performance of the method proposed in this paper. Some comparisons are given as well.

$s(t)$ is the LFM signal during 1s, with sample frequency $f_s = 1000$, amplitude $A_0 = 1$, center frequency $f_0 = 0$ and modulated frequency $k = 100$. $s_{tfd}(t)$ is the measured signal with a joint time-frequency offset $t_d = 0.4s, f_d = 10Hz$. $s_{td}(t)$ and $s_{fd}(t)$ represent the single time delay and frequency offset version of $s(t)$, respectively. The waveforms of original signal $s(t)$, joint time-frequency offset signal $s_{tfd}(t)$ and $s_{tfd}(t)$ under noise environment for $SNR = -10dB$ are given in Fig. 1.

Figure 2 (a) shows the amplitude spectrums of $s(t)$, $s_{td}(t)$, $s_{fd}(t)$, $s_{tfd}(t)$ in Fourier domain. According to these basic properties of FT, the time delay can not be detected using the Fourier spectrums. Then two steps are used: First, the frequency offset $f_d = 9.9900$ is obtained using correlation between the Fourier spectrums of $s(t)$ and $s_{tfd}(t)$; Second, the time delay is $t_d = 0.3$ taking the frequency estimator into $s_{tfd}(t)$ and using Eq. 8. The Mean Squared Error (MSE) is 0.0711. Since the t_d is obtained after f_d , this method increases the error. In Fig. 2 (b), the Fourier spectrums with noise are given when $SNR = -10dB$. From this figure, it is hard to clarify the signals and noise. The method based on the FT is influenced by noise severely. LCT spectrums of $s(t)$, $s_{td}(t)$, $s_{fd}(t)$ and $s_{tfd}(t)$ without noise are shown in Fig. 2(c) and according spectrums with noise are in Fig. 2(d)

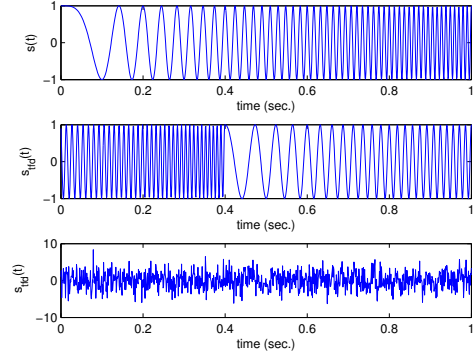


Fig. 1. Waveforms of $s(t)$, $s_{tfd}(t)$ and noisy $s_{tfd}(t)$ ($SNR=-10$ dB).

when LCT parameters $a/b = -2\pi k$. Signals and noise can be obviously detected since signals are concentrated in such LCT domains. Under $SNR=-10dB$, the values for single time delay and frequency offset are $t_d = 0.3996$, $f_d = 9.990$, and $MSE=0.0071$. However, $t_d = 0.4021$, $f_d = 10.0524$, and $MSE=0.0371$ are in FRFT domain when α equals to $-\text{arccot}(2\pi k)$. MSE varies over different SNRs for FRFT and LCT is given in Fig. 3.

Lastly, LCT is used to detect the joint time-frequency offset in the noisy environment. Here, six different values of a/b are chosen in order to solve the parameters t_d and f_d out, which are 3.7321, 1.7321, 1.0000, 0.8391, 0.5774, 0.1826. According to Eq. 12, the parametric correlation attains its maximum when $u^A = at_d + bf_d$, which is a line in the time-frequency plane. Thus, intersection point in the plane exactly is joint time-frequency offset needed to detect. Six different lines are shown in the time frequency plane in Fig. 4. They are denoted by $l_i, i = 1, 2, 3, 4, 5, 6$.

By combining the each two lines, the interaction points can be obtained. The results are listed in the Table 1. Column 1 shows the numbers of interacted lines; Column 2 and Column 3 are the interaction points (t_d, f_d) can be found in the time frequency plane. Column 4 gives the MSE. At the last row in this table, the mean of $t_d = 0.4163$, $f_d = 9.9842$. The MSE for the mean values is 0.0161. For α modulus correlation, the MSE of joint time-frequency offset is 0.1886. Using the new method can also achieve the detection of joint time-frequency offset detection.

TABLE I
DETECTION RESULTS USING NEW CORRELATION METHODS

Lines	Time delay t_d	Frequency offset f_d	MSE
1,2	0.4578	9.8998	0.0818
1,4	0.4400	9.9670	0.0367
1,6	0.4328	9.9943	0.0235
2,4	0.3993	10.0012	0.0010
2,6	0.3998	10.004	0.0028
3,5	0.3994	10.007	0.0050
4,5	0.4011	9.997	0.0023
5,6	0.4001	10.0003	0.0021
Mean	0.4163	9.9842	0.0161

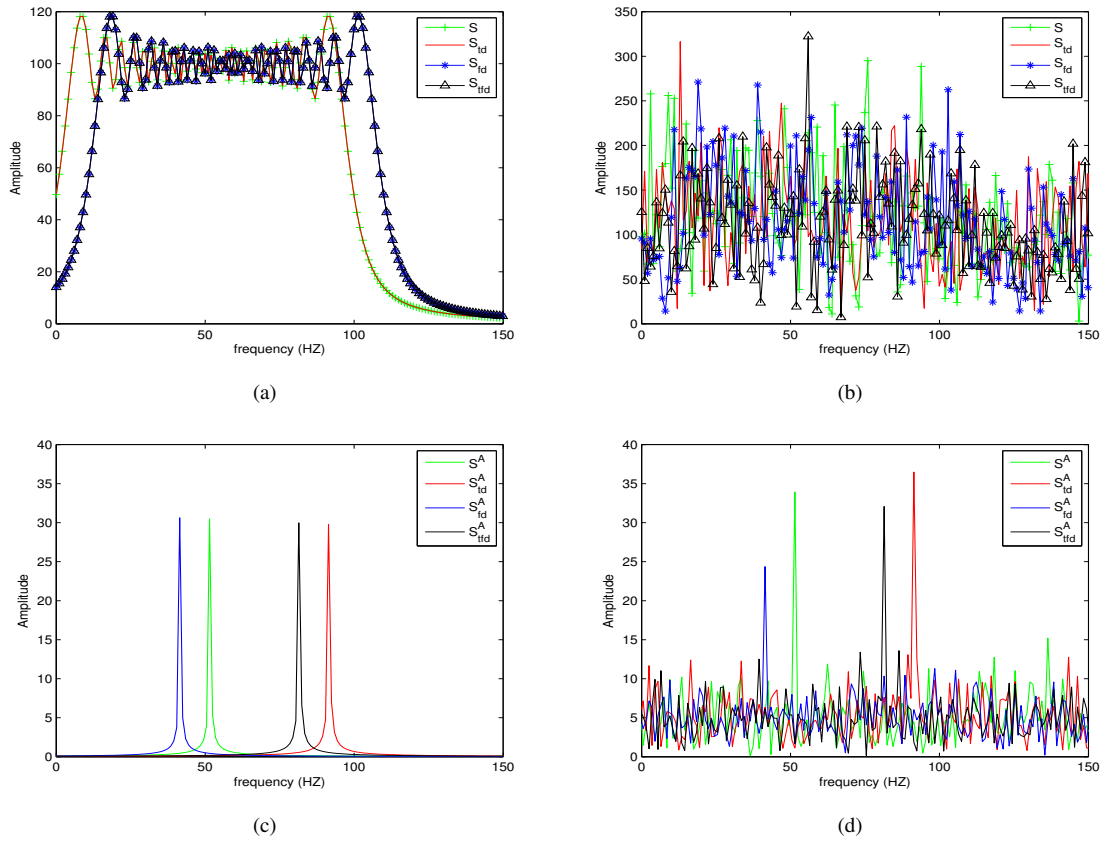


Fig. 2. Amplitude spectrum of $s(t)$, $s_{td}(t)$, $s_{fd}(t)$, $s_{tfd}(t)$ in Fourier domain and LCT domain.(a) In Fourier domain without noise. (b) In Fourier domain with noise (SNR=-10dB). (c) In LCT domain without noise. (d) In LCT domain with noise (SNR=-10dB)

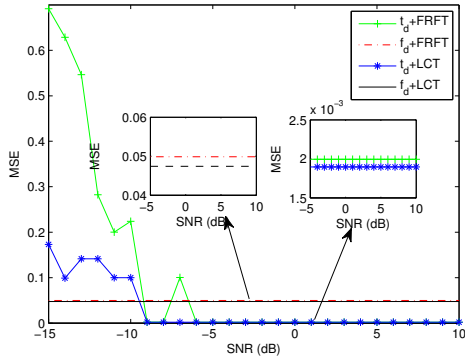


Fig. 3. MSE based on FRFT and LCT

V. CONCLUSION

This paper proposes a parametric correlation to detect the joint time-frequency offset. It is a generalization of the techniques in the frame of correlation based on FT and FRFT to detect time delay and frequency offset. According to the basic properties of LCT, the maximum of parametric correlation contains the information about time and frequency, which is a line structure in the time frequency plane. Based on the

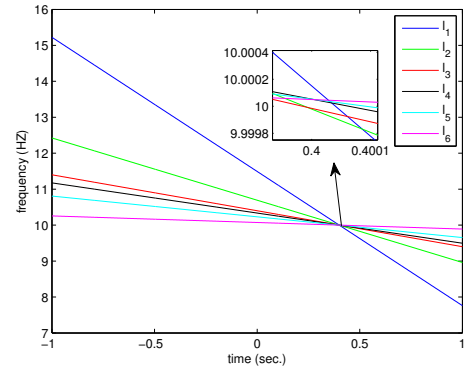


Fig. 4. Different lines in the time-frequency plane using new method

freedom of parameters matrix in LCT, some lines can intersect in the plane. Just to find out these intersected point, the joint time-frequency offset can be detected directly. Compared with common correlation, this method can detect the time delay and frequency offset simultaneously.

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